Kogut–Susskind strong coupling: stability, finite-volume bounds and history estimates

The Kogut–Susskind Hamiltonian has a volume-uniform strong-coupling gap by Yarotsky's established local-perturbation theorem. For $SU(3)$ the theorem also covers the translation-invariant, gauge-invariant few-link perturbations and link-local electric deformations specified below. Its constants and the location of the resulting target region are existential.

At a fixed volume the physical electric-loop gap has an explicit two-sided min–max estimate. Continuous-time histories give additional conditional overlap bounds and restricted shape counts. They supply **no explicit volume-uniform KS threshold**: the old values $g^2 = 79,388$ and the associated uniform rates were withdrawn because late roots, general shapes and normalized true-vacuum correlations were not controlled. The corrections are incorporated at each estimate below. Wilson-transfer thresholds concern a different operator and remain in their own proof.

All gaps below are energies, with physical unit $\hbar c/a$. Positivity along a weak-coupling scaling trajectory, cutoff removal and physical reconstruction are separate obligations; positivity at every bare coupling is an optional stronger property of a lattice action.

Hamiltonian and the local stability theorem

On a periodic cubic spatial lattice $\Lambda = (\mathbb{Z}/N_s\mathbb{Z})^3$, with links $\mathcal{E}$ and plaquettes $\mathcal{P}$, use

$$H = \frac{\hbar c}{a}\left[\frac{g^2}{2}\sum_{\ell}(-\Delta_\ell) + \frac{2}{g^2}\sum_{p}(N - \operatorname{Re}\operatorname{tr} U_p)\right].$$

For $SU(N)$, $\operatorname{tr} T^a T^b = \delta^{ab}/2$ and $C_2(R_{\min}) = (N^2-1)/(2N)$. Write $\varepsilon_0 = g^2 C_2/2$ in lattice energy units. Each site space $\mathcal{H}_x = L^2(G)^{\otimes 3}$ contains its three outgoing links, with preferred constant vector Ω_x. The Peter–Weyl isotypic projections give a product partition of unity, and

$$h_x = C_2^{-1}\sum_{i=1}^{3}(-\Delta_{(x,x+e_i)})$$

is a function of that partition, vanishes on Ω_x and is at least one on its orthogonal complement.

Yarotsky, *CMP* **261** (2006) 799, §1 and Theorem 1 (**passage read**), applies to these possibly infinite-dimensional site spaces and a translation-invariant finite-range bounded perturbation. For range Λ_0 there are positive $\beta_*(3,\Lambda_0)$ and γ such that a normalized local perturbation norm at most β_* leaves a simple ground state and gap at least γ, uniformly in periodic volume. It also gives thermodynamic ground states and clustering for a fixed consistent interaction. The bounded case uses relative coefficient $\alpha = 0$; no numerical value of β_* is supplied by the statement.

Assign each of the three plaquettes $p(x,i,j)$, $i<j$, to x. After division by ε_0, their positive potential has local norm at most

$$\beta(g) = \frac{24N}{g^4 C_2} = \frac{48N^2}{g^4(N^2-1)}.$$

Its sites lie in $x + \{0,1\}^3$. Thus, for $N_s \geq 2$ and $\beta(g) \leq \beta_*(3,\{0,1\}^3)$,

$$\Delta_{a,L}^{\text{phys}} \geq \Delta_{a,L} \geq \gamma\frac{g^2 C_2}{2}\frac{\hbar c}{a}.$$

This is a direct hypothesis check, not a new proof of the borrowed theorem. For a compact connected group with a specified faithful unitary representation of dimension d_R, the same sufficient estimate is $g^4 \geq 24 d_R/(C_2\beta_*)$. For the unperturbed elliptic Hamiltonian, positivity

improvement gives a unique positive ground state, and gauge invariance of that vector follows from uniqueness and normalization (finite-lattice proof). The physical subspace reduces H and contains its ground state, proving gap inheritance.

The displayed sufficient g_0 is bounded as $N \to \infty$ at fixed g. At fixed 't Hooft coupling $\lambda = g^2 N$, its condition instead grows with N. Neither fact transports this theorem to $g(a) \to 0$.

Bounded local perturbations and the existential target region

In units $\hbar c/a = 1$ let

$$H = H'_E - g^{-2} \sum_p w_p - \sum_j d_j, \qquad w_p = \operatorname{tr} U_p + \operatorname{tr} U_p^\dagger, \quad \|w_p\| \le 6.$$

Assume all of the following for $SU(3)$:

- $H'_E = \sum_{\ell,R} e'(R) P_R^\ell$ is link-local and translation invariant, $e'(\text{triv}) = 0$, and $e'(R) \ge \varepsilon'$ for every nontrivial R, where $\varepsilon' = (1 - \eta_E) 2g^2/3$, $0 \le \eta_E < 1$.
- Each d_j is bounded, self-adjoint and gauge invariant, on a connected set of at most q links. The family is translation covariant and $\sum_{j \ni \ell} \|d_j\| \le \eta$ for every link.
- $q \ge 1$, $N_s \ge 2q + 1$, and $\Lambda_q = \{-q, \dots, q\}^3$.

Let $n_x(j)$ count support links based at x and distribute each term by

$$\delta_x = \sum_j \frac{n_x(j)}{|\operatorname{supp} d_j|} d_j.$$

Then $\sum_x \delta_x = \sum_j d_j$, $\|\delta_x\| \le 3\eta$, and its support lies in $x + \Lambda_q$: connectedness and at most q links give the stated radius. Dividing by ε' gives an on-site electric part with unit gap and local perturbation

$$\phi_x = -\frac{1}{\varepsilon' g^2} \sum_{i<j} w_{p(x,i,j)} - \frac{\delta_x}{\varepsilon'}, \qquad \|\phi_x\| \le \frac{1}{1 - \eta_E} \left(\frac{27}{g^4} + \frac{9\eta}{2g^2} \right).$$

The difference between 54 in the uncentered $SU(3)$ estimate above and 27 here comes from subtracting the magnetic scalar constant; both are sufficient bounds with their specified ranges.

Stability theorem. With $\beta_q^* = \beta_*(3, \Lambda_q)$ and the corresponding $\gamma_q > 0$, the condition

$$\frac{27}{g^4} + \frac{9\eta}{2g^2} \le (1 - \eta_E)\beta_q^*, \quad \text{equivalently} \quad \eta \le \frac{2(1 - \eta_E)\beta_q^*}{9} g^2 - \frac{6}{g^2}$$

implies a simple gauge-invariant ground state and

$$\Delta_{a,L}^{\text{phys}} \ge \Delta_{a,L} \ge \gamma_q (1 - \eta_E) \frac{2g^2}{3} \frac{\hbar c}{a}.$$

Here η is a numerical local norm in lattice energy units. The target region opens at $g^4 \ge 27/[(1 - \eta_E)\beta_q^*]$.

Proof. The preceding grouping verifies Yarotsky's hypotheses. A unique normalized ground ray transforms under $SU(3)^\Lambda$ by a continuous one-dimensional character. The Lie algebra is semisimple, so its derivative vanishes; connectedness makes the character trivial. The ground state is therefore physical even though general d_j need not preserve positivity. The reducing physical subspace inherits the lower bound. □

The bound is uniform for a volume-dependent family satisfying the same hypotheses. A thermodynamic state assertion additionally uses one consistent infinite-lattice interaction, or the appropriate convergent family. The borrowed theorem does not automatically cover nontranslation-invariant perturbations, disconnected supports of unbounded diameter, or an arbitrary nonlocal diagonal electric energy. Its range dependence is existential; a 2^{-q} law is not part of this theorem.

Physical electric loops and exact finite-volume bounds

For the unperturbed $SU(3)$ operator write $H = H_E - W + 6P/g^2$, with $P = 3N_s^3$ and

$$E(\sigma) = \frac{g^2}{2}\sum_\ell C_2(r_\ell) \geq \varepsilon|S(\sigma)|, \quad \varepsilon = 2g^2/3, \qquad \|W\| \leq 6P/g^2.$$

Gauss's law excludes a vertex with only one excited incident link. Nonabelian invariant networks may branch; they need not be unions of flux loops. Every nonempty finite graph of minimum degree at least two contains a cycle. For $N_s \geq 4$ its length is at least four. A plaquette character realizes four fundamental links and is orthogonal to the electric vacuum, so the physical electric gap is exactly 4ε. Shorter periodic lattices need separate winding counts; the unrestricted electric gap is only ε.

Min–max bounds the shifts of each of the two lowest levels by $\|W\|$:

$$\left(\frac{8}{3}g^2 - \frac{12P}{g^2}\right)\frac{\hbar c}{a} \leq \Delta_{a,L}^{\text{phys}} \leq \left(\frac{8}{3}g^2 + \frac{12P}{g^2}\right)\frac{\hbar c}{a}.$$

At each fixed volume the ratio to $g^2\hbar c/a$ tends to $8/3$. The sufficient $g^4 \geq 9P$ gives lower coefficient $4/3$ but grows with volume. It cannot establish a uniform threshold.

The abelian comparison and a finite-volume bound

For $U(1)$ use $H_E = (g^2/2)\sum n_\ell^2$ and magnetic potential $(2/g^2)\sum(1-\cos\theta_p)$. The same four-edge argument gives

$$\left(2g^2 - \frac{4P}{g^2}\right)\frac{\hbar c}{a} \leq \Delta_{a,L}^{\text{phys}} \leq \left(2g^2 + \frac{4P}{g^2}\right)\frac{\hbar c}{a}, \qquad N_s \geq 4.$$

The local stability theorem also gives a volume-uniform strong-coupling region for $U(1)$. The weak sides differ: $SU(3)$ has perturbative $dg^2/d\log a = 2b_0g^4 + O(g^6)$, $b_0 = 11/(16\pi^2)$, whereas a pure continuum photon has no such self-interaction. Guth, *PRD* **21** (1980) 2291, and Fröhlich–Spencer, *CMP* **83** (1982) 411, are the retained metadata-level sources for the compact-lattice weak-coupling massless phase. Running alone proves neither controlled arrival at the target region nor continuum spectral transfer.

Continuous-time histories and the corrected overlap criterion

This is an additional candidate expansion, not the proof of the stability theorem. It also allows the more general diagonal physical H'_E commuting with exact-support projections P_S, with $E'(\sigma) \geq \varepsilon'|S|$. Assume $N_s \geq \max(4, 2q+1)$. Center the bounded d_j in the electric vacuum; scalar subtraction can at most double the quoted local norm η. Without centering, remote vacuum insertions need not carry energy damping.

Duhamel's ordered expansion of $e^{-t(H'_E - W - D)}$ inserts plaquettes or d_j terms. Resolving P_S between them gives histories $S_0 = S_n = \emptyset$ with insertion operators X_k, gaps $x_k = \tau_{k+1} - \tau_k$ and

$$A(\gamma) = \sum_{k=1}^{n-1}|S_k|x_k, \qquad |w(\gamma)| \leq \prod_k \|X_k\| e^{-\varepsilon' A(\gamma)}.$$

Amplitudes can have either sign. Labels within an exact support are already summed in the operator norm. A plaquette changes only its four links and has at most 16 next-support outcomes; a q-link insertion has at most 2^q. An irreducible physical history has $|S_k| \geq 4$ between insertions.

Disjoint link factors and chronological interleavings motivate a hard-core history gas, but a complete construction must still specify connected components and insertion-endpoint compatibility. The following overlap criterion is **conditional on such a factorizing gas and its**

majorant. For an established measure-space gas, the KP theorem used here is Poghosyan–Ueltschi, §2, Assumptions 1–2 and Theorem 2.1 (passage read), including its finite-volume/time integrability premise.

Set $a(\gamma) = n + \lambda A$, $\delta' = \varepsilon' - \lambda > 0$. The tilted majorant has insertion weights multiplied by e. Let T_ℓ be the total excited duration and r_ℓ the number of maximal excitation intervals on link ℓ. Then

$$\sum_\ell T_\ell = A, \qquad \sum_\ell r_\ell \le \max(4,q)n.$$

An interval of length x' in another history can overlap these intervals under translations of total measure at most $T_\ell + r_\ell x'$. The multiplicity r_ℓ is essential.

Let Q' majorize the sum over **all** connected shapes reaching a test link, including a sum over marked internal intervals, after integrating $e^{-\delta'|S_k|x_k}$ on the positive orthant. Its length-marked version is at most $Q'/(4\delta')$, since

$$\int_0^\infty e^{-\delta'|S|x}dx = \frac{1}{\delta'|S|}, \qquad \int_0^\infty x e^{-\delta'|S|x}dx \le \frac{1}{4\delta'}\frac{1}{\delta'|S|}.$$

Therefore the overlap integral is at most

$$Q'A(\gamma) + \frac{Q'\max(4,q)}{4\delta'}n(\gamma),$$

and sufficient conditional inequalities are

$$Q' \le \lambda, \qquad Q' \le \frac{4\delta'}{\max(4,q)}.$$

For unperturbed plaquette histories this is $Q_\lambda \le \lambda, \delta$, where explicitly

$$Q_\lambda = \sup_{\ell_0} \sum_{\text{shapes reaching } \ell_0} (n-1)h^n \prod_{k=1}^{n-1} (\delta|S_k|)^{-1}, \qquad h = 6e/g^2.$$

The old $Q_\lambda \le 4\delta$ lost the endpoint multiplicity four. Neither version proves that the actual all-root majorant is small.

Restricted shape counts and their numerical status

Anchor the **first** insertion at the test link and require every later insertion to touch the current support. For the unperturbed gas four initial plaquettes give $F = 4h = 24e/g^2$. Later choices are at most $4|S| \cdot 16$, whose factor $|S|$ is canceled by the preceding time integral. For the perturbed gas the same calculation gives

$$F' = 24e/g^2 + e\eta 2^q, \qquad u' = \frac{384e/g^2 + e\eta 2^q}{\delta'},$$

$$B'_\lambda \le F' \sum_{m\ge 1} m(u')^m = \frac{F'u'}{(1-u')^2} \le 4F'u' \quad (u' \le 1/2).$$

This is a valid restricted count. It includes repeated insertions. It omits histories first reaching the test link later and pieces created remotely and subsequently bridged; hence B'_λ does not bound Q'.

For general connected insertion graphs, assign consecutive link intervals to a spanning tree T. A surviving energy inequality is

$$\sum_{e\in T} |\tau_{i(e)} - \tau_{j(e)}| \le A(\gamma), \qquad e^{-\delta A} \le \prod_{e\in T} e^{-\delta|\tau_i - \tau_j|}.$$

Each insertion has at most eight predecessor/successor link slots. This does not count the required root, interval marks and time-length multiplicities. The former $u_{\mathrm{gen}} = 24u$ also allowed only three child plaquettes per parent link, excluding the repeated plaquette that creates and annihilates a loop. Changing that factor to four would give a formal $32u$, still without a proved marked count. Neither is a certified KP parameter.

For $\eta = \eta_E = 0$, write $\lambda = \varepsilon(1-\theta)$, $\delta = \varepsilon\theta$, $0 < \theta < 1$. Then

$$u = \frac{576e}{\theta g^4}, \qquad g^4 \geq \frac{1152e}{\theta}, \quad g^8 \geq \frac{82944e^2}{\theta \min(\theta, 1-\theta)}$$

are sufficient for the displayed restricted B_λ inequalities only.

θ	sufficient integer g^2 for those restricted inequalities	candidate 4λ, not a proved Hamiltonian rate
1/2	80	$4g^2/3$
1/4	112	$2g^2$
1/10	177	$2.4g^2$

The former adjacent 79 rounds down: $79^2 < 2304e$. For the withdrawn formal $24u$ alone, $24u = 13824e/(\theta g^4)$ and $24u \leq 1/2$ requires $g^4 \geq 27648e/\theta$. Integers 388, 549, 867 satisfy that **algebraic condition** at the three listed θ values; the old middle 548 was rounded down. None is a uniform gap threshold. The old additional $g^8 \geq 1.47 \times 10^7/[\theta(1-\theta)]$ inherited the unproved count.

For the perturbed restricted count, $\theta = 1/2$, $\eta_E = 0.1$ give $\lambda = \delta' = 0.3g^2$. Requiring the formal B' majorant to meet the overlap conditions yields

$$e\eta 2^q \leq 0.15g^2 - 384e/g^2, \qquad e\eta 2^q \leq 0.6g^2/\max(4,q) - 24e/g^2.$$

g^2	$q = 4$ restricted η	$q = 8$ restricted η	binding condition for $q = 8$
100	0.104	0.00655	first
200	0.569	0.0210	second
400	1.31	0.0428	second

Values are rounded down, in lattice energy units; the first insertion's 2^q multiplicity is included. At large g^2, $\theta \leq 1/2$, the formal slope is $4\theta(1-\eta_E)g^2/[3\max(4,q)e2^q]$. This is a property of the restricted count, not the existential theorem's range dependence. The former suggestion to divide tolerances by 24 at the same couplings also fails algebraically: $24u' \leq 1/2$ would require $e\eta 2^q \leq 0.00625g^2 - 384e/g^2$, with no nonnegative solution unless $g^4 \geq 61440e$. None of these tables locates the proved target region.

From normalized true-vacuum correlations to a gap

Let $\psi_0 > 0$ be the normalized true finite-volume ground state and $f_A = (A - \langle A\rangle_0)\psi_0$, for bounded gauge-invariant multiplication observables, including whole-lattice supports. If

$$0 \leq \langle f_A, e^{-t(H-E_0)} f_A\rangle \leq C_A e^{-mt} \quad (t \geq 0)$$

holds for every such A, then $\Delta^{\mathrm{phys}} \geq m\hbar c/a$ in this dimensionless time convention. Indeed positive spectral mass below $m - \zeta$ would contradict the exponential bound. The f_A are dense

in the physical vacuum complement: truncate the gauge-invariant ratio f/ψ_0 and then center. Bounded spectral projections extend their vanishing from that dense set. Infinite volume additionally needs a named vacuum GNS representation and dense observable domain.

This spectral lemma's hypothesis has not been proved by the history count. For true-vacuum-centered A, the rank-one ground projection gives only

$$\frac{\langle \Omega_0, e^{-TH} A^* e^{-tH} A e^{-TH} \Omega_0 \rangle}{\langle \Omega_0, e^{-(2T+t)H} \Omega_0 \rangle} \longrightarrow \langle \psi_0, A^* e^{-t(H-E_0)} A \psi_0 \rangle.$$

A proof of decay needs a normalized connected insertion bound uniform in boundary distance T and volume. Individual four-link histories alone do not show that each correlation cluster spans area at least $4t$. Thus $m = 4\lambda$, a uniform limiting coefficient four in electric-gap units, and any asserted $O(g^{-4})$ relative correction remain unproved here.

Time-discretized comparison and its missing activity bound

For the uncentered $SU(3)$ normalization the local perturbation is $\beta \le 54/g^4$. The earlier proof-extraction estimate assumed support entropy c^n, $c = 80e$, per-site activity ρ^n, and $\sum_{n\ge1}(ce\rho)^n \le 1$, hence $\rho \le 1/(160e^2)$. It further assumed

$$\rho_J = e^{-t_0}, \qquad \rho_I = 2et_0\beta e^{64t_0}.$$

The 64 is the proposed shadow volume $|\{0,1\}^3|^2$; the insertion factor minimizes $2\alpha e^{t_0\beta/\alpha}$ at $\alpha = t_0\beta$. Taking $t_0 = 7.1 \ge \log 160 + 2$ would require

$$\beta \le \frac{e^{-454.4}}{160e^2\, 2e \cdot 7.1}, \qquad \log\beta \lesssim -465, \qquad \log g_0^2 \simeq \tfrac{1}{2}(\log 54 + 465) \simeq 234.5.$$

The resulting order 10^{102} is a **heuristic extraction**, not a numerical conclusion of Yarotsky's theorem: conversion of full configuration weights and multiplicities into the assumed support-site bound is missing. Nor is e^{64t_0} an unavoidable cost of all discretized methods. The alternative illustrative coordination estimate $g^4 \gtrsim 54 \cdot 12e$, suggesting g^2 of order 40–70, has no applicable Kirkwood–Thomas or Datta–Kennedy constant proved here; those sources were read at metadata level only.

Exact low-energy reductions and the physical scale

Suppose a specified scheme gives $H(a)$ and $H_{\mathrm{eff}}(2a)$ whose spectra, with multiplicities and a common scalar shift, agree below $E_c(a) > E_0(a)$. Then

$$\Delta(H(a)) \ge \min\{\Delta(H_{\mathrm{eff}}(2a)), E_c(a) - E_0(a)\}.$$

If a finite sequence reaches the stability theorem at $a_s = 2^s a_{\mathrm{UV}}$, in the same full or physical sector at every step, iteration gives

$$\Delta(H(a_{\mathrm{UV}})) \ge \min\left\{\gamma_q(1-\eta_{E,s})\frac{2g_s^2}{3}\frac{\hbar c}{a_s}, \min_{k<s}[E_c(a_k) - E_0(a_k)]\right\}.$$

Uniform positive physical margins and controlled a_s are additional hypotheses as $a_{\mathrm{UV}} \to 0$; merely having finitely many positive margins at each cutoff is insufficient. An actual generated interaction must satisfy the theorem's locality and covariance premises. Reconstruction, nontrivial spectral weight and the physical limit stay separate (spectral obligations).

The one-loop orientation $n \simeq (g_{\mathrm{weak}}^{-2} - g_{\mathrm{strong}}^{-2})/0.0966$ gives about 21 doublings from $g^2 = 1/2$ to a hypothetical order-10^2 endpoint, or order 10^3 when the initial inverse coupling is 100. These are extrapolations; the intermediate region and its corrections are uncontrolled. The small-volume Feshbach bridge uses a different basis and small parameter and retains its ultraviolet caveat.

Consequence for STATE

The local stability theorem, exact finite-volume bounds, conditional history criterion and restricted numerical tests now share one maintained treatment. Only the first supplies volume uniformity. An explicit continuous-time threshold requires the missing all-root count and normalized true-vacuum correlation estimate; a physical continuum gap additionally requires cutoff-uniform transport and reconstruction along the weak trajectory.